

CMP416 Network Analysis

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Abstract

Network forensic analysis is the process of identifying potential security faults on a network by examining all aspects, virtual and physical, of a network. In an ideal state a tester would have access to a plethora of information, from physical machines to routers and server rooms. However, oftentimes testers are left with minimal resources, as is the case in this examination.

The challenge in network forensics lies in making the most of a microscopic amount of information. Log files, browser lists, momentary network trafficking can be enough to diagnosed issues on their own, allowing for testers to bring forth vulnerabilities without the physical requirements. Due to factors such as encryption, machine learning, storage limitations as well as any potential physical damages to vulnerable systems, network forensics can often be limited (Focus, 2025). The importance of making the most of information in forensics will be demonstrated in the examination below, where testers utilize system and network logs to properly identify and examine network vulnerabilities amongst systems.

For the sake of this examination all traffic associated with the construction of the Virtual Network involved Sean Sturley was discredited as unintended artefacts. In a real world forensic examination this would be accounted for, however due to the nature of the traffic being generated by the instructors home device the focus on this examination was related to the times associated with the Mimikatz mimicked infection.

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1 EXECUTIVE SUMMARY

1.1 NETWORK OVERVIEW

The testers received a package containing folders related to four different host machines as well as a syslog file and a network traffic file. The testers were given information on the network via a json file, indicating that the subnet of the network was 192.168.4.0/24 and the gateway was 192.168.4.1. The date and time of collection occurred at 2025-11-23 T02:58:23.077174. The four hosts machines have been identified as *WORKSTATION-758*, *PC-491*, *LAPTOP-142*, and *DESKTOP-500*. These host machine folders contain several folders of interest, such as browser information, logs, memory files, network files, and registry folders. Included within these folders is a malware folder, identified as a potential vulnerability to examine later on in testing. From a surface level observation of the network folders for each machine the IP and MAC addresses associated with each machine as laid out as follows:

- | | | | |
|---|-----------------|-------------------|--------------|
| - | DESKTOP-500 | 08-00-27-8A-A9-8B | 192.168.4.13 |
| - | LAPTOP-142 | 08-00-27-90-55-D9 | 192.168.4.12 |
| - | PC-491 | 08-00-27-A2-D9-20 | 192.168.4.11 |
| - | WORKSTATION-758 | 08-00-27-5F-EB-A6 | 192.168.4.10 |

The hierarchy of the network will be determined through testing, properly identifying the permissions and roles of each system on the network.

1.2 AIMS

The aims of this network examination are as follows:

- Perform a thorough network analysis of the CMP416 network.
- Identify any malicious activity possibly present on the network and subsequently the machines at hand.
- Identify possible security concerns within the network configuration and permission hierarchies.
- Establish an infection timeline.
- Provide suggestions and solutions to any identified security vulnerabilities.

1.3 SECURITY CONCERNS

This project aims to follow the security principles set forth by the ACPO. All data granted to the user will be examined through read only tools that ensure data integrity is kept throughout the project. Any and all access to private files shall be documented and only examined when necessary to diagnose the networking infections behaviour and damages. Furthermore, all files created by third party examination tools shall be provided and taken account for whilst abiding by Scottish law.

2 INVESTIGATION PLAN

2.1 OVERVIEW OF PROCEDURE

The investigation will follow a rudimentary network forensic methodology based heavily on the efforts of Splunk. Laid out below are the steps that have been taken during the investigation:

- Identification
- Examination
- Analysis
- Presentation

Due to the nature of the information provided to the investigators, the data preservation and collection steps have been omitted as the network information has already been provided to the investigators beforehand. During the Identification process, the examiners will utilize the network information provided alongside third party tools such as Wireshark, tshark, and tcpdump to develop an identity to each user on the network during the time of the incident. Once the users have been properly identified, a process examination will be conducted on each machine to identify possible targets of interest. With the Wireshark analysed and the infected hosts identified, the logs left behind will be examined for any linked malicious activity at times of interest. Lastly whilst utilizing strings, the testers will search the memory of these hosts for further malicious activity to add to the comprehensive report.

Once these examinations have taken place, a timeline of infection will be constructed to properly illustrate the path of infection and how users were compromised. The vulnerabilities discovered will be discussed and described in detail as the method of discovery is laid out for the network consultant. During this conclusion, network security recommendations will be provided to ensure future vulnerabilities are mended to prevent threat actors penetrating the network.

3 EVIDENCE AND ANALYSIS

3.1 WIRESHARK NETWORK ANALYSIS

The testers began the examination by inputting the network traffic into Wireshark, prompting an error message. This error is mainly associated with unsensitized packets, as seen in Figure 3.1, indicating that the traffic was pulled before packets completed their handshake(tshark, 2025).

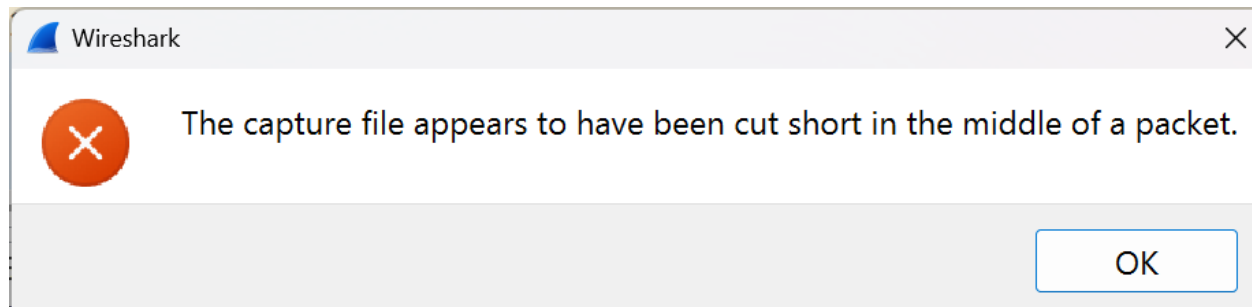


Figure 3.1 Screenshot of Wireshark notification relating to network_capture.pcap.

The network_capture was searched individually by source ip, associating each filter with traffic relating to each host address. Within packet communication with PC-491 and WORKSTATION-758 there were several calls to wdc.microsoft.com, indicating that Windows Defender was flagged during the allotted runtime. Searching through Wireshark with the filter (**dns.qry.name == "wdcp.microsoft.com"**) as well as (**dns.qry.name == "wdcp.microsoft.com"**) revealed that only these hosts communicated with Microsoft Defender, indicating that a threat might have infiltrated these systems. The time on these communications was between 533.045887 and 1005.5797, with the first instance being on the WORKSTATION-758 host.

No.	Time	Source	Destination	Protocol	Length	Info
2040	752.103501	192.168.4.1	192.168.4.11	DNS	197	Standard query response 0xa781 A wdc.microsoft.com CN
1974	730.426380	192.168.4.1	192.168.4.11	DNS	200	Standard query response 0xacb7 A wdc.microsoft.com CN
2609	948.485464	192.168.4.1	192.168.4.11	DNS	197	Standard query response 0xbd2b A wdc.microsoft.com CN
2688	982.897212	192.168.4.1	192.168.4.11	DNS	197	Standard query response 0xc9c8 A wdc.microsoft.com CN
1684	654.507509	192.168.4.1	192.168.4.10	DNS	197	Standard query response 0xdad1 A wdc.microsoft.com CN
1613	633.164800	192.168.4.1	192.168.4.10	DNS	200	Standard query response 0xe0ce A wdc.microsoft.com CN
1487	588.183427	192.168.4.1	192.168.4.10	DNS	197	Standard query response 0xe1bb A wdc.microsoft.com CN
2555	926.512983	192.168.4.1	192.168.4.11	DNS	199	Standard query response 0xe83b A wdc.microsoft.com CN
2078	775.058203	192.168.4.1	192.168.4.11	DNS	202	Standard query response 0xef7f A wdc.microsoft.com CN

Figure 3.2 Screenshot of the dns.qry.name filter in Wireshark.

Below is a table of each protocol in the pcap file alongside the frequency of the protocol. Utilizing the protocol hierarchy statistics feature in Wireshark, the testers were able to confirm that all packets were transferred through ethernet upon the network.

ARP x547	ICMPv6 x3	SSDP x1
BROWSER x2	IGMPv3 x2	Syslog x17

DHCP x11	MDNS x90	TCP x1701
DNS x373	NTP x1	DHCPv6 x2

3.2 NETWORK TEXT FILE ANALYSIS

When analysing the provided text files, a few notable points of interest were examined. Notably, when examining the systeminfo files and comparing their contents between machines, WORKSTATION-758 systeminfo appeared blank for the testers. This indicated that something actively wiped the systeminfo from the provided documentation. Furthermore, the netstat.txt for the LAPTOP-142 indicated access and activity assigned to port 123. Port 123 is generally responsible for network time protocols and is particularly vulnerable to denial of service (DDoS) attacks.

Analysing the arp.txt files indicates that LAPTOP-142 had two physical connections linked to IP multicast addresses. This could indicate that the LAPTOP-142 host had a higher level of privileges due to the MAC addresses permissions indicated here from this arp command output. The indications that they are able to send out multicast messages through these channels indicates to the tester this device might have higher privileges.

When analysing the process listings on each machine the LAPTOP-142 lsass process was taking 20-25x the amount of CPU as compared to the other hosts. When analysing Google for any suspicious lsass.exe activity, the search informed the analysts of possible Mimikatz activity. Mimikatz is an open source application that allows users to view and save authentication credentials(Buckbee, 2025). The testers made a note of the potential infiltration from this point and looked for footprints related to Mimikatz activity for the remainder of the testing process.

3.3 EVENT VIEWER

The analyst moved further utilizing the information gathered from earlier testing to examine the possibility of Mimikatz infiltration. Mimikatz is known for utilizing logon access, and when examining an event log one can expect to find several calls to log on/log off of specific services. Notably, this is generally associated with Event ID 4648, which relates to log on request on Microsoft machines. When examining the Security log of LAPTOP-142, as seen in figure 3.3.1, the testers discovered a plethora of calls for this Event ID, indicating several authentication attempts from the suspected machine. These calls began at 21:13:08 on November 15th, 2025, however, activity really began five days later. Expanding the Event log to filter for more Mimikatz authentication related event IDs, the testers discovered more of the same, as can be seen in Appendix A. November 20th, 2025 seemed to be the dated of infection on the device, where the time began 15:54:55.

Information	20/11/2025 15:55:03	Microsoft Windows se...	4648	Logon
Information	22/11/2025 21:00:45	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:27:58	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:34:00	Microsoft Windows se...	4648	Logon
Information	22/11/2025 22:39:35	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:36:11	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:56:17	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:42:47	Microsoft Windows se...	4648	Logon
Information	22/11/2025 21:09:34	Microsoft Windows se...	4648	Logon
Information	22/11/2025 20:55:12	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:28:17	Microsoft Windows se...	4648	Logon
Information	22/11/2025 20:55:23	Microsoft Windows se...	4648	Logon
Information	21/11/2025 12:11:23	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:42:33	Microsoft Windows se...	4648	Logon
Information	21/11/2025 12:11:23	Microsoft Windows se...	4648	Logon
Information	21/11/2025 12:11:29	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:33:52	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:42:21	Microsoft Windows se...	4648	Logon
Information	21/11/2025 12:11:43	Microsoft Windows se...	4648	Logon
Information	20/11/2025 16:42:21	Microsoft Windows se...	4648	Logon

Figure 3.3.1 Event Viewer of several logon attempts on LAPOTOP-142.

Examining all the events from each device revealed an SID S-1-5-21-1975775866-2975552529-277365765-1002. S-1-5-21 SIDs are associated with users on the network, indicating a user is flagging these events against all machines. The events associated with this SID indicate logins were attempted and succeeded from this user. Figure 3.3.2 displayed Powershell commands that were ran on the WORKSTATION-758 host, these commands searched the provided registries for the SID. From running these commands, the testers were able to determine that the labuser ID associated with the admin of the network was utilized to login to each of these machines.

```

Administrator: Windows PowerShell
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\WINDOWS\system32> reg load HKLM\TempSOFTWARE "C:\Users\ultra\OneDrive\Documents\CMP416\WORKSTATION-758\registry\SOFT
FTWARE"
The operation completed successfully.
PS C:\WINDOWS\system32> reg query "HKLM\TempSOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1975775866
-2975552529-277365765-1002" /v ProfileImagePath
HKEY_LOCAL_MACHINE\TempSOFTWARE\Microsoft\Windows NT\CurrentVersion\ProfileList\S-1-5-21-1975775866-2975552529-277365765
-1002
ProfileImagePath    REG_EXPAND_SZ      C:\Users\labuser
PS C:\WINDOWS\system32> reg unload HKLM\TempSOFTWARE
The operation completed successfully.
PS C:\WINDOWS\system32>

```

Figure 3.3.2 Screenshot of Powershell commands inputted on the systems.

3.4 HxD

Inspecting the Malware folders of each machine revealed that WORKSTATION-758 contained a unique malware.exe. Utilizing a third party hex editor HxD the malware file revealed behaviour indicating the process was a spoof of a popular authentication tool, Mimikatz. This spoof matches information found earlier in testing from the Event Viewer as well as the provided Network Text Files. Below is a screenshot of the HxD result of the Mimikatz executable, displaying a message that was left by the threat actor.

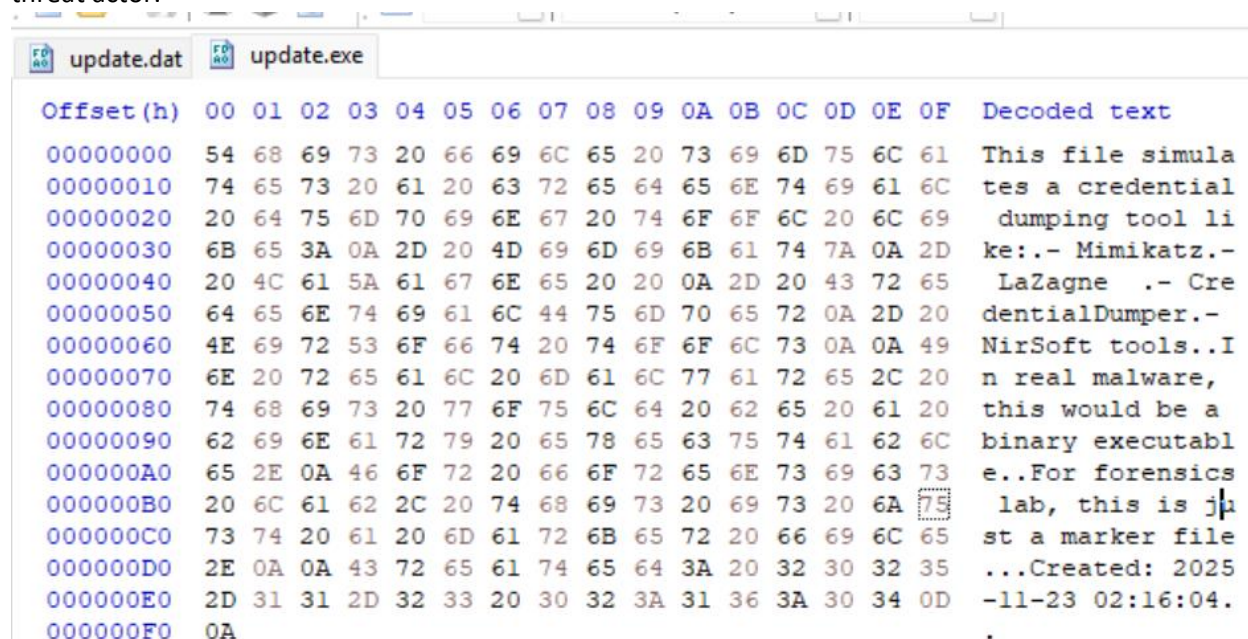


Figure 3.4.1 Screenshot of the HxD of update.exe.

Furthermore, a dat file associated with the malware package was left, and when inputted into the HxD file this revealed several spoofed credentials from the systems, matching with the used credential events found in the Event Viewer observations. The labuser admin user was seemingly compromised, leaving Chrome passwords, LinkedIn credentials, and more exposed. Below is a table for all the exposed credentials from the update.dat file, indicating the entire network was compromised. The most valuable note left from this dat file was the time of attack, 2025-11-23 02:16:03. This will help the testers to lay out an appropriate timeline for the events of the attack.

Application	Used ID	Password
Mail	j.smith@company.com	P@sswOrd123!
LinkedIn	john.smith	LinkedIn2023!
Github	jsmith-dev	GitHubToken_abc123
Portal	j.smith	CompanyP@ss2023
Bank	jsmith75	SecureBank#456
Outlook	j.smith@company.com	Office365Pass!
Teams	j.smith@company.com	Office365Pass!
Reddit	jsmith_reddit	RedditPass789
Twitter	@johnsmith	TwitterP@ss456

VPN	Jsmith	VPN_P@ssw0rd!
<pre> === Credential Harvesting Results === Timestamp: 2025-11-23 02:40:24 Target: WORKSTATION-758 User: labuser [BROWSER PASSWORDS] Chrome Password Store: https://mail.google.com - j.smith@company.com : P@ssw0rd123! https://www.linkedin.com - john.smith : LinkedIn2023! https://github.com - jsmith-dev : GitHubToken_abc123 https://portal.company.com - j.smith : CompanyP@ss2023 https://www.bank.com - jsmith75 : SecureBank#456 Edge Password Store: https://outlook.office.com - j.smith@company.com : Office365Pass! https://teams.microsoft.com - j.smith@company.com : Office365Pass! Firefox Password Store: https://www.reddit.com - jsmith_reddit : RedditPass789 https://twitter.com - @johnsmith : TwitterP@ss456 Total Browser Passwords Extracted: 9 [WINDOWS CREDENTIALS] Credential Manager: WindowsLive:target=virtualapp/didlogical - john.smith@outlook.com Domain:target=WORKGROUP - WORKSTATION-758\labuser Generic:target=Company VPN - jsmith : VPN_P@ssw0rd! SAM Database Hashes (NTLM): </pre>		

Figure 3.4.1 Screenshot of memory.bin showing passwords.

Moreover, there were several accounts, Administrator and User, compromised. The administrator account compromised had a hash ID of r:500:aad3b435b51404eeaad3b435b51404ee:8846f7eaae8fb117ad06bdd830b7586c. the r:500 revealing that the account was an administrator, indicating that there was lateral movement upon the network. The lowest privileged exposed account from this dat file was a user RID of 1001:aad3b435b51404eeaad3b435b51404ee:b4b9b02e6f09a9bd760f388b67351e2b. This ID determined to be the labuser user on the network from further powershell investigation, utilizing similar commands as was seen in Figure 3.3.2. With these credentials, the malware package could infiltrate all aspects of the network, leaving all devices compromised. During the Wireshark analysis, it can be safely assumed that all traffic on these devices after the date linked on the .dat file can be associated with these malware package as long as they are linked to the labuser or admin credentials.

3.5 MEMORY.BIN

Lastly, to quell any suspicions the testers examined the memory.bin file by converting the contents into a Strings-based output. Utilizing the windows download for strings, a memory.txt file was created showcasing all available text that was within the memory of the WORKSTATION-758 machine. This memory.txt file was utilized as an input for a PowerShell add-on known as Select-String, which functions very similarly to grep in Linux. The following command was inputted into an administrator level Powershell window:

Select-String -Path "C:\Users\ultra\Downloads\Strings\memory.txt" -Pattern "malware" > "C:\Users\ultra\Documents\malware.txt"

This command resulted into a malware.txt file that flagged Windows Defender on the host examiners PC, as can be seen in Figure 3.5.1. This showcases a potential hazardous string within the contents of the memory.bin on the infected host. Inputting this same potentially malicious string into VirusTotal showcases a similar result to Windows Defender, as seen in Figure 3.5.2.

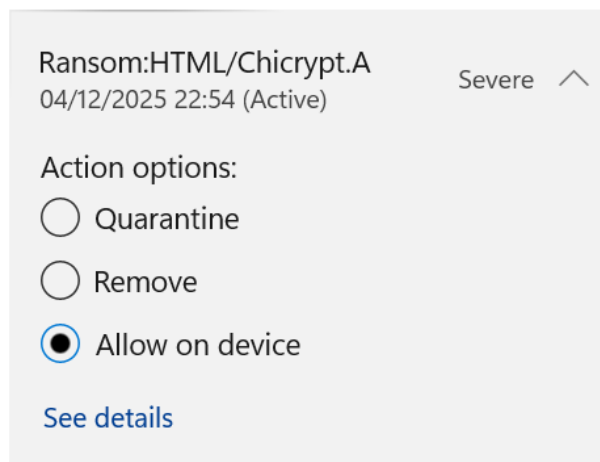


Figure 3.5.1 Windows Defender Flag from malware.txt result.

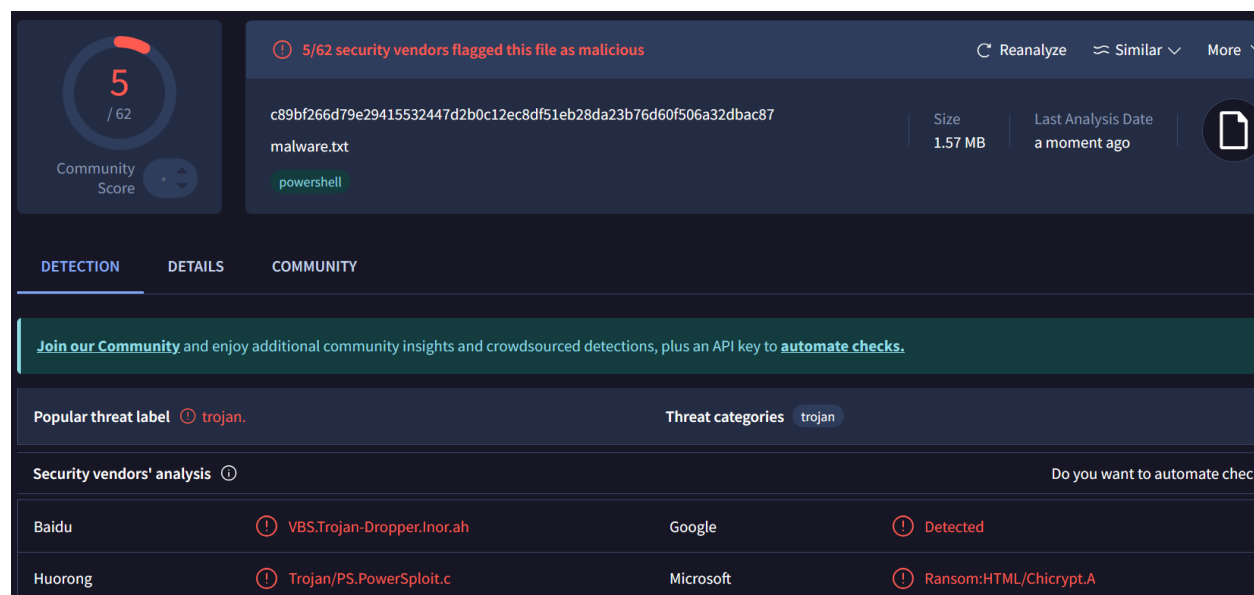


Figure 3.5.2 VirusTotal Result flagging the malware.txt file as malicious.

These anti-viral programs seem to refer to the string as potentially the same viral family, Chicrypt. According to sources such as malpedia and MalwareBytes, Chicrypt is a pseudonym for the Chimera malware family, which was a prolific ransomware family in mid 2010s that utilized doxxing threats in combination of their extortion efforts (Fraunhofer, 2015). Further research into the matter showcased that Chimera ransomware utilized a deviant of the Mimikatz program showcased to Microsoft in their

attack methods, extracting the key methods from the software and holding host information hostages (CyCraft, 2020).

4 FINDINGS AND CONCLUSIONS

4.1 SUSPICIOUS ACTIVITY

From the information noted whilst examining the network as well as the hosts involved, the examiners can determine that a Mimikatz deviant was infiltrated onto the WORKSTATION-758 host machine. This could be observed through the malware executable located within this host machines files, containing the output of infiltrated data as well. This data can be observed through the Network Analysis and Event Viewers provided, following the methods in which this data crawler exfiltrated logins for several services on the network. Furthermore, utilizing administrative level Windows PowerShell commands the examiners were able to uncover the RID and SID of several spoofed users, notably the Admin account as well as the labuser account. This information showcased lateral movement from low privilege accounts to higher privilege accounts. These accounts were then used to fully infected each machine as a PowerShell History was left on each machine showcasing the commands that the malware used to “rat” credentials. These credentials can be found in the table on section 3.4, showcasing all logins found in the network during the time of the attack.

4.2 MALICIOUS PACKAGES

Update.exe mimics a similar package, Mimikatz, in an attempt to view and save login credentials(Buckbee, 2025). Mimikatz works by utilizing Windows own password storage methods against the user. These methods, such as hashing, are often vulnerable as in previous versions Windows would accept the hash as equivalent to the passwords, leaving their practice valueless. In recent iterations of Windows, pass the key works much of the same way, exfiltrating administrative level credentials by utilizing their unique key related to their passwords as inputs. For Windows, the remaining methods of attack relate to the Kerberoast services, generating a “golden ticket” that gives the malware administrative permissions when granted.

4.3 FURTHER VULNERABILITIES

Beyond the Mimikatz infection, there were several other possible vulnerabilities discovered on the network unrelated to the attack. Notably, LAPTOP-142 was found to have an open port 123. This port in particular has been linked to DDoS attacks as it relates to Network Time Protocols. This service has been utilized for clock synchronization over a network, however, when overwhelmed this port can be responsible for entire network collapse. Furthermore, several authentications amongst the users were found to have loose credentials, with passwords oftentimes barely meeting the minimal credentials for services. It is best to practice more complex password generation than what is required as well as two factor authentication practices to best protect your users.

4.4 INFECTION TIMELINE

Nov 23, 2025 02:16:03.000000000 GMT – Infection Begins, Trojan ran on app.exe

Nov 23, 2025 02:40:24.000000000 GMT – Harvesting Result Finished, as seen in Figure 3.4.1

Nov 23, 2025 02:50:03.307897000 GMT Standard Time – first Windows Defender flag

Nov 23, 2025 02:57:55.841715000 GMT Standard Time – last Windows Defender flag

5 RECOMMENDATIONS

5.1 COUNTERMEASURES

The analysts recommended a full revaluation of firewall rules on the systems. Firewall rules would be able to detect any lateral movement attempts and prevent such attacks from continuing beyond the scope of one host. Furthermore, restricting administrative permissions to a few seldom host machines that are tech savvy enough to ensure Mimikatz deviants would not be downloaded from a social engineering attack. Disabling password caching would prevent Mimikatz from utilizing Windows services that save credentials to scrub logins from users. On all host machines certain debugging permissions should be prevented as they allow for services to act as developers on Windows machine and manipulate otherwise obfuscated data. Finally, a comprehensive User Training demonstration would be recommended to ensure that each host has the properly knowledge to protect themselves as well as the network from any malicious actors. From emails to in person, ensure proper social engineering prevention is crucial when protecting a network.

5.2 FUTURE WORK

Given more time allotted the examiners would love to disassemble the malware package found on the network and properly examine the functionality within utilizing tools such as Ghidra. These tools could help increase the understanding of the vulnerabilities utilized on the network as well as the fundamental processes that allow for a package to exfiltrate data the way update.exe has. Continuing, an Autopsy report would be useful to further examine leaks possibly missed during the examination, giving a more comprehensive view of the network to the examiners. The testers would also with more time want to continue scrubbing the browser data with different yara rules developed as well as an examination of the syslog files that were left unreported due to time constraints.

The testers recommend a follow-up session two months down the line to ensure that the network is secured from this point forward. This test will ensure that the proper recommendations have been implemented sufficiently to ensure a future attack is unlikely.

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